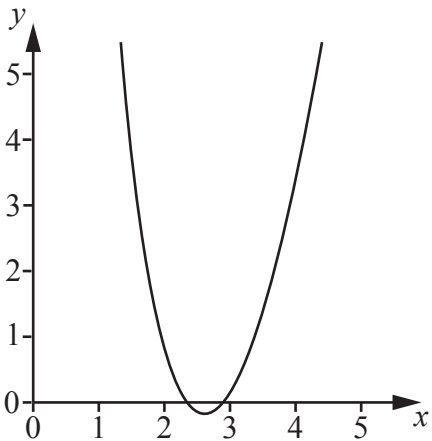
6 The diagram shows the curve with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 13x - 34\ln(x) - 1.5$ for $x > 0$.



(a) A student attempts to locate a root of $\mathrm{f}(x) = 0$ by using a spreadsheet to calculate the values of $\mathrm{f}(x)$ for positive integer values of x.

Explain why the student's method may lead to the conclusion that $\mathrm{f}(x) = 0$ has no roots. **[1]**

(b) Determine the exact value of the x-coordinate of the turning point of the curve. **[2]**

6(a)	
6(b)	